

General Concepts of Physics and Methods of Solving Physics Problems

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1 General Concepts of Physics

1.1 Physical System

Physical system

A *physical system* is a collection of physical objects, or even a single physical object.

1.2 State of a Physical System

The *mechanical state of a single particle* is given by **six quantities**:

- 1) three position coordinates: (x, y, z) ;
- 2) three components of the particle's momentum: (p_x, p_y, p_z) .

Equivalently, the state is one point (x, y, z, p_x, p_y, p_z) in a six-dimensional space; a system of N particles therefore needs $6N$ quantities.

1.3 Interaction

Objects in a physical system are interrelated both with each other and with external objects. This interrelationship manifests itself in the *interaction* of physical objects.

Types of interaction:

- 1) Gravitational
- 2) Electromagnetic
- 3) Weak nuclear
- 4) Strong nuclear

Interaction	Range	Approx. relative strength
Gravitational	infinite	$\sim 10^{-38}$
Electromagnetic	infinite	$\sim 10^{-2}$
Weak nuclear	$\sim 10^{-18}$ m	$\sim 10^{-6}$
Strong nuclear	$\sim 10^{-15}$ m	~ 1

(Order-of-magnitude values, quoted for two protons.)

Key idea

Interaction changes the position or state of the system.

1.4 Physical Law

Physical law

A necessary and stable interrelation (dependence) between physical quantities is expressed by a *physical law*.

Every law has two aspects that must be known:

- 1) the **conditions** for its application;
- 2) the **method** for its application.

2 Idealization of a Physics Problem

Formulated physics problem

A *formulated* physics problem is a problem concerned with a pure, idealized phenomenon. The author severs the links between the given physical phenomenon and other phenomena.

Example 2.1: Projectile

A projectile is fired by a gun at an angle $\alpha = 45^\circ$ to the horizon with initial speed $v_0 = 600 \text{ m/s}$. Find the horizontal distance to which the projectile will be propelled. Ignore air drag.

Solution. With $x = v_0 \cos \alpha t$ and $y = v_0 \sin \alpha t - \frac{1}{2}gt^2$, landing ($y = 0$) occurs at $t = 2v_0 \sin \alpha / g$, so

$$R = \frac{v_0^2 \sin 2\alpha}{g} = \frac{(600)^2 \cdot 1}{9.8} \text{ m} \approx 3.7 \times 10^4 \text{ m} \approx 37 \text{ km}.$$

This problem is **formulated and idealized**. Besides the condition stated by the author (ignore air drag), there are other *tacitly assumed* conditions, for example: uniform gravitational field (flat Earth), no wind, Earth's rotation neglected, projectile treated as a point particle, launch and landing at the same height. We can lift these conditions one by one to make the problem more complex.

2.1 Two Ways of Idealization

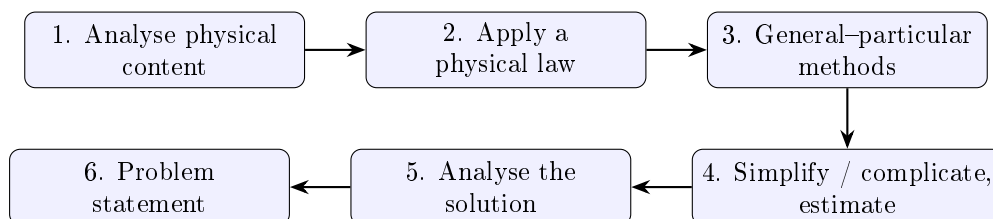
- **Introduction of idealized physical objects** (point particle, rigid body, ideal gas, ...).
- **Neglect of non-essential interactions and processes** (friction, air drag, ...).

2.2 Three Stages of Solving a Problem

- 1) **Physical** stage,
- 2) **Mathematical** stage,
- 3) **Analysis of the solution** — this stage helps in solving other problems.

3 The General Method

- 1) Analysis of the physical content of a problem.
- 2) Application of a physical law.
- 3) General-particular methods: kinematics, dynamics, conservation laws, methods of calculating physical fields, and the method of differentiation and integration.
- 4) Simplification and complication methods, and estimation.
- 5) Analysis of the solution.
- 6) Problem statement.



The following sections treat three of these in detail: the **DI method** (step 3), the **estimate method** (step 4) and the **problem statement method** (step 6).

4 The Method of Differentiation and Integration (DI)

The DI method is used to deal with the *limits of applicability of physical laws*: a law that holds only for a small (elementary) part of a system is applied in its **differential form** to that part, and the contributions are then added using the **superposition principle**.

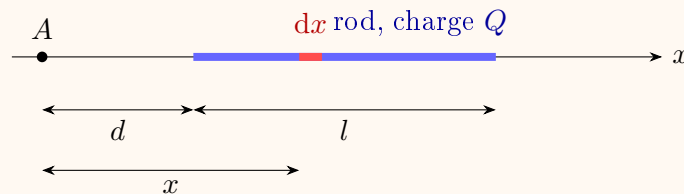
Principle

If a law has the form $K = L M$, apply it to an elementary part,

$$dK = L(M) dM \quad \Rightarrow \quad K = \int_{M_1}^{M_2} L(M) dM.$$

Example 4.1: Potential of a charged rod

A rod of length $l = 1$ m carries a uniformly distributed charge $Q = 10^{-12}$ C. Find the potential at the point A on the rod's axis at distance $d = 1$ m from its nearest end.



Point-charge estimate (valid only if $d \gg l$):

$$\varphi \approx \frac{kQ}{d} = \frac{9 \times 10^9 \cdot 10^{-12}}{1} \text{ V} = 9 \times 10^{-3} \text{ V}.$$

DI method. Coulomb's potential law holds for a point charge, so apply it to an element dQ at distance x from A . The linear charge density is constant, $\frac{dQ}{dx} = \frac{Q}{l}$ (in general $dQ = \lambda(x) dx$). Then

$$d\varphi = \frac{k dQ}{x} = \frac{kQ}{l} \frac{dx}{x} \quad \Rightarrow \quad \varphi = \frac{kQ}{l} \int_d^{d+l} \frac{dx}{x} = \frac{kQ}{l} \ln\left(1 + \frac{l}{d}\right).$$

For $d = l = 1$ m: $\varphi = 9 \times 10^{-3} \ln 2 \approx 6.2 \times 10^{-3} \text{ V}$.

5 The Estimate Method

The estimate method is used to solve **non-specific and non-standard** problems. Its principle is **simplification and complication**. It consists of:

- 1) estimating orders of magnitude;
- 2) comparing orders of magnitude.

Rule for discarding terms

The terms of greatest order are isolated; terms whose order is lower than the highest order by **at least two** are discarded.

Example 5.1: Change of gas mass in a cylinder

The change of gas mass is

$$\Delta m = \frac{VM(P_1T_2 - P_2T_1)}{RT_1T_2},$$

with $V = 9 \text{ L}$, $M = 2 \times 10^{-3} \text{ kg/mol}$, $P_1 = 5.2 \times 10^6 \text{ Pa}$, $T_1 = 296 \text{ K}$, $P_2 = 5 \times 10^4 \text{ Pa}$, $T_2 = 285 \text{ K}$, $R = 8.31 \text{ J/(mol K)}$. Estimate the order of magnitude of Δm .

Solution. Convert to SI and round:

$$V \approx 10^{-2} \text{ m}^3, \quad M = 2 \times 10^{-3} \text{ kg/mol}, \quad R \approx 8 \text{ J/(mol K)}, \\ P_1 \approx 5 \times 10^6 \text{ Pa}, \quad P_2 \approx 5 \times 10^4 \text{ Pa}, \quad T_1 \approx T_2 \approx 3 \times 10^2 \text{ K}.$$

Since $T_1 \approx T_2 = T$, the formula simplifies:

$$\Delta m \approx \frac{VM(P_1 - P_2)}{RT}.$$

Since $P_1 \gg P_2$, P_2 can be ignored:

$$\Delta m \approx \frac{VMP_1}{RT} = \frac{10^{-2} \cdot 2 \times 10^{-3} \cdot 5 \times 10^6}{8 \cdot 3 \times 10^2} \text{ kg} \approx 4 \times 10^{-2} \text{ kg}.$$

The exact calculation gives $\Delta m = 3.8 \times 10^{-2} \text{ kg}$.

Example 5.2: Gravity versus electric force between two protons

$$\frac{F_g}{F_e} = \frac{Gm^2/r^2}{Q^2/(4\pi\epsilon_0 r^2)} = \frac{Gm^2 \cdot 4\pi\epsilon_0}{Q^2} \approx 8 \times 10^{-37} \sim 10^{-36}.$$

Data: $G \approx 6.7 \times 10^{-11} \text{ N m}^2/\text{kg}^2$, $m \approx 1.67 \times 10^{-27} \text{ kg}$, $Q = 1.6 \times 10^{-19} \text{ C}$, $4\pi\epsilon_0 \approx 1.1 \times 10^{-10} \text{ F/m}$. Gravity is utterly negligible compared with the electric force at atomic scales.

Example 5.3: Does the Earth or the Sun attract the Moon more strongly?

$$\frac{F_E}{F_S} = \frac{M_E r_S^2}{M_S r_E^2} \approx \frac{6 \times 10^{24}}{2 \times 10^{30}} \left(\frac{1.5 \times 10^{11}}{4 \times 10^8} \right)^2 \approx 0.4,$$

with $M_E \approx 6 \times 10^{24} \text{ kg}$, $M_S \approx 2 \times 10^{30} \text{ kg}$, $r_E \approx 4 \times 10^8 \text{ m}$ (Earth–Moon), $r_S \approx 1.5 \times 10^{11} \text{ m}$ (Sun–Moon). Hence the **Sun attracts the Moon about 2.5 times as strongly as the Earth does.**

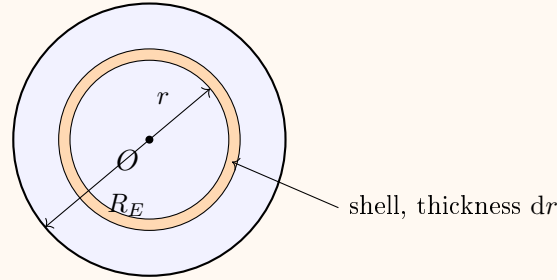
Remark. The Moon still orbits the Earth because the Sun accelerates the Earth and the Moon almost equally; only the small *difference* of the Sun's pull matters for the Moon's motion relative to the Earth.

Estimating a physical phenomenon

Estimating a phenomenon amounts to finding the fundamental law governing it and then numerically calculating the order of magnitude of the respective physical quantity. Estimate problems are often non-standard.

Example 5.4: Pressure at the Earth's centre

Treat the Earth as a ball of uniform density ρ and radius R_E (a shell of mass dm at radius r is pulled inward by the mass $M(r)$ inside it, and $p = F/A$).



Mass inside radius r and mass of the shell:

$$M(r) = \frac{4}{3}\pi r^3 \rho, \quad dm = 4\pi r^2 \rho dr.$$

Force on the shell and the resulting pressure increment:

$$dF_g = \frac{G dm M(r)}{r^2} = \frac{G \cdot 4\pi r^2 \rho dr \cdot \frac{4}{3}\pi r^3 \rho}{r^2}, \quad dp = \frac{dF_g}{4\pi r^2} = \frac{4\pi G \rho^2}{3} r dr.$$

Integrating from r to R_E :

$$P(r) = \int_r^{R_E} dp = \frac{2\pi}{3} G \rho^2 (R_E^2 - r^2).$$

At the centre:

$$P(0) = \frac{2\pi}{3} G \rho^2 R_E^2 \approx 1.6 \times 10^{11} \text{ Pa}.$$

Standard atmospheric pressure is 10^5 Pa, so the pressure at the centre of the Earth exceeds it by **six orders of magnitude**.

Note: here ρ is the Earth's mean density, $\rho = 3M_E/(4\pi R_E^3)$. The real density increases towards the core, so the true central pressure is higher (about 3.6×10^{11} Pa), but the order of magnitude is right.

6 The Problem Statement Method

Idealization is followed by formulation. Then the restrictions and additional conditions are **lifted** step by step, which turns a simple problem into a more complex one.

Example 6.1: Induced current: simplify, then complicate

A planar loop of area $A = 1 \text{ m}^2$ and resistance $R = 1 \Omega$ is placed in a magnetic field perpendicular to its plane, with

$$B(t) = B_0 - \frac{\alpha t^2}{2A}, \quad B_0 = 10 \text{ T}, \quad \alpha = 10^{-1} \text{ T m}^2/\text{s}^2.$$

Induced EMF:

$$\varepsilon_1 = -\frac{d\Phi}{dt} = -A \frac{dB}{dt} = \alpha t.$$

(Only dB/dt matters; B_0 drops out.)

Simple problem (self-induction EMF $\varepsilon_{si} = -L \, dI/dt$ ignored for small L):

$$I_1 = \frac{\varepsilon_1}{R} = \frac{\alpha t}{R} = 0.1 \, t \, \text{A} \quad (t \text{ in seconds}).$$

Lifting the restriction (large L). Kirchhoff's second rule gives

$$\varepsilon_1 - L \frac{dI}{dt} = IR \quad \implies \quad \alpha t - L \frac{dI}{dt} = IR.$$

With the initial condition $I = 0$ at $t = 0$, the solution is

$$I(t) = \frac{\alpha t}{R} - \frac{\alpha L}{R^2} \left(1 - e^{-(R/L)t} \right).$$

Checks: for $L \rightarrow 0$ this returns $I = \alpha t/R$; for $t \ll L/R$ it gives $I \approx \alpha t^2/(2L)$; for $t \gg L/R$ the current follows $\alpha t/R$ with a delay L/R .

6.1 Classification of Formulated Problems

Type	Uses	Used for training
Elementary	One physical law.	Quick recall ("fast" thinking).
Standard	Common knowledge, a standard approach and a standard method of applying it.	Analysing a system and the method of applying a law.
Non-standard	The system of equations is not closed; requires guesses.	Creativity.

Know the law *and* how to apply it

We must not only know the law but also how to apply it. This is trained by **standard problems**, where you analyse the physical content of a problem and the method of applying the necessary physical law.

Original problem

An *original problem* is a non-standard problem in which the role of a wild guess is much greater than the role of ordinary methods ($\gg$); it requires creativity.

7 Quick Revision Summary

- **State** of one particle: 6 quantities (x, y, z, p_x, p_y, p_z) . **Interactions:** gravitational, electromagnetic, weak, strong.
- A **physical law** is a necessary, stable relation between quantities; know its *conditions* and its *method* of application.
- **Idealization:** introduce idealized objects; neglect non-essential interactions. Solve in three stages: physical, mathematical, analysis.
- **DI method:** $dK = L(M) \, dM \Rightarrow K = \int L(M) \, dM$ (differential form + superposition).
- **Estimate method:** keep the leading orders, drop terms smaller by two or more orders of magnitude; ask which law governs, then compute its order of magnitude.

- **Problem statement method:** idealize, formulate, then lift restrictions one by one.
- **Problem types:** elementary (one law), standard (known approach), non-standard (open system of equations, needs guesses).